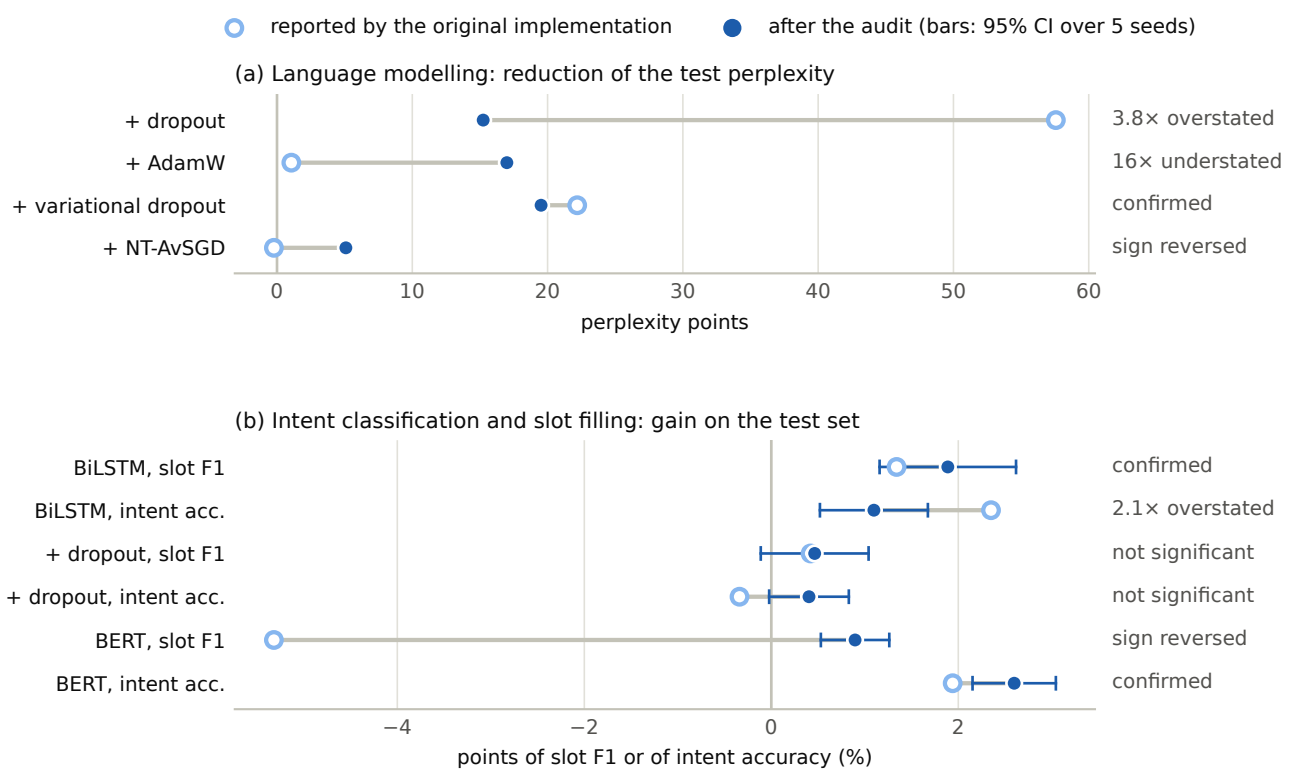
reported by the original implementation
after the audit (bars: 95% CI over 5 seeds)
(a) Language modelling: reduction of the test perplexity
+ dropout
+ AdamW
+ variational dropout
+ NT-AvSGD
3.8× overstated
16× understated
confirmed
sign reversed
0
10
20
30
40
50
60
perplexity points
(b) Intent classification and slot filling: gain on the test set
BiLSTM, slot F1
BiLSTM, intent acc.
+ dropout, slot F1
+ dropout, intent acc.
BERT, slot F1
BERT, intent acc.
confirmed
2.1× overstated
not significant
not significant
sign reversed
confirmed
−4
−2
0
2
points of slot F1 or of intent accuracy (%)